

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1399

Kentucky Dam 138 kV
February 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1399. The Interconnection Customer proposes a 390.5 MW solar plus storage facility interconnecting at the Kentucky Dam 138 kV substation in the Virginia area. The request was received on January 30, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 349 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,971,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: KENTUCKY DAM – STOCK ISLAND 138kV
Contingency: KENTUCKY DAM – DEERFIELD BEACH 138kV
Loading: 174.4% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 4.7%
2. Facility: KENTUCKY DAM – DEERFIELD BEACH 138kV Line
Contingency: STOCK ISLAND – KENTUCKY DAM SUBSTATION 138kV
Loading: 169.9% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 4.6%
3. Facility: WHITE MESA – DEERFIELD BEACH 138kV L
Contingency: KENTUCKY DAM – STOCK ISLAND 138kV
Loading: 161.8% (Rating: 230 MVA)
DFAX: 100.0%
Pre-project loading: 8.1%
4. Facility: BROADFORD – WHITE MESA 138kV
Contingency: KENTUCKY DAM – STOCK ISLAND 138kV
Loading: 138.9% (Rating: 259 MVA)
DFAX: 100.0%
Pre-project loading: 12.0%
5. Facility: OROVADA – TOLSON 138kV line
Contingency: 43RD & S. LAND – CEMENT PLANT 138kV
Loading: 135.9% (Rating: 226 MVA)
DFAX: 38.7%
Pre-project loading: 69.0%

6. Facility: TOLSON – US ORD DEPOT-HERLING 138kV
Contingency: 43RD & S. LAND – CEMENT PLANT 138kV
Loading: 126.4% (Rating: 253 MVA)
DFAX: 38.7%
Pre-project loading: 66.5%
7. Facility: PALMER – N BRISTOL 138kV
Contingency: E BERKSHIRE – WOLF HILLS ENERGY 345kV
Loading: 112.3% (Rating: 265 MVA)
DFAX: 23.4%
Pre-project loading: 77.8%
8. Facility: PALMER – WOLF HILLS ENERGY 138kV
Contingency: WOLF HILLS ENERGY – E BERKSHIRE 345kV
Loading: 108.8% (Rating: 288 MVA)
DFAX: 23.4%
Pre-project loading: 77.0%
9. Facility: HYAK – FRIENDSHIP 138kV
Contingency: STA 11 – STOCK ISLAND 138kV
Loading: 107.3% (Rating: 237 MVA)
DFAX: 29.8%
Pre-project loading: 58.2%
10. Facility: NASSAU – INSKIP 138kV
Contingency: LOST NATION – NO TROY SUBSTATION 138kV
Loading: 107.2% (Rating: 188 MVA)
DFAX: 25.2%
Pre-project loading: 54.8%
11. Facility: DAFFIN GIN – WHITING AVE 138kV line
Contingency: FRIENDSHIP – VACHERIE 138kV L
Loading: 106.2% (Rating: 167 MVA)
DFAX: 14.4%
Pre-project loading: 72.6%
12. Facility: WHITING AVE – TORRINGTON TERMINAL 138kV Line
Contingency: VACHERIE – FRIENDSHIP 138kV
Loading: 105.0% (Rating: 178 MVA)
DFAX: 14.4%
Pre-project loading: 73.4%
13. Facility: INSKIP – BLUEFIELD AVENUE 138kV
Contingency: LOST NATION – NO TROY 138kV
Loading: 103.6% (Rating: 187 MVA)
DFAX: 25.2%
Pre-project loading: 51.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KENTUCKY DAM – STOCK ISLAND 138kV	Rebuild line with higher-capacity conductor	\$5,726,000	100% (\$5,726,000)
KENTUCKY DAM – DEERFIELD BEACH 138kV Line	Rebuild line with higher-capacity conductor	\$5,749,000	100% (\$5,749,000)
WHITE MESA – DEERFIELD BEACH 138kV line	Rebuild line with higher-capacity conductor	\$27,196,000	100% (\$27,196,000)
BROADFORD – WHITE MESA 138kV	Rebuild line with higher-capacity conductor	\$2,106,000	100% (\$2,106,000)
OROVADA – TOLSON 138kV	Rebuild line with higher-capacity conductor	\$18,728,000	32% (\$5,891,000)
TOLSON – US ORD DEPOT-HERLONG 138kV	Rebuild line with higher-capacity conductor	\$10,139,000	32% (\$3,189,000)
PALMER – N BRISTOL 138kV Line	Rebuild line with higher-capacity conductor	\$4,342,000	37% (\$1,613,000)
PALMER – WOLF HILLS ENERGY 138kV	Reconductor line	\$776,000	37% (\$288,000)
HYAK – FRIENDSHIP 138kV line	Reconductor line	\$1,352,000	33% (\$444,000)
NASAU – INSKIP 138kV	Reconductor line	\$3,037,000	54% (\$1,639,000)
DAFFIN GIN – WHITING AVE 138kV line	Reconductor line	\$1,432,000	28% (\$400,000)
WHITING AVE – TORRINGTON TERMINAL 138kV L	Reconductor line	\$5,223,000	28% (\$1,459,000)
INKIP – BLUEFIELD AVENUE SUBSTATION 138kV	Reconductor line	\$2,356,000	54% (\$1,271,000)
	Total allocated to project		\$56,971,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.